

Sundareswari Thiyagarajan

M.S. in Electrical and Electronics Engineering

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RESEARCH PROFILE

M.S. Student at Inha University working on multimodal speech AI, speaker verification, silent speech recognition, and deployment-aware edge AI. Published peer-reviewed research in Symmetry and Electronics (MDPI Journal) and authored work presented at IEEE BigComp 2026. Experienced in hybrid attention/state-space architectures, audio-visual fusion, quality-aware verification, quantized LLM deployment, and benchmarking on NVIDIA Jetson AGX Xavier.

EDUCATION

M.S. in Electrical and Electronics Engineering, Inha University Feb 2025 - Expected Feb 2027

Intelligent Embedded Systems Laboratory, Incheon, Republic of Korea | Advisor: Prof. Deok-Hwan Kim

- Research focus: speaker verification, silent speech recognition, multimodal learning, and embedded edge AI.

B.E. in Computer Science and Engineering, Anna University Aug 2019 - Jun 2023

Tiruchirappalli, Tamil Nadu, India

RESEARCH EXPERIENCE

Graduate Researcher, Intelligent Embedded Systems Laboratory Feb 2025 - Present

Department of Electrical and Computer Engineering, Inha University

- Develop speaker-verification systems using TDNN, BiLSTM, multi-head attention, state-space modeling, ArcFace/AAM-Softmax, and quality-aware decision support.
- Designed an audio-visual verification pipeline with a ResNet-18 lip encoder, gated fusion, and shared Conformer; achieved 3.419% EER and 0.342 minDCF on VoxCeleb evaluation.
- Built HySV, a hybrid attention/state-space framework evaluated for accuracy and deployment efficiency using EER, minDCF, FLOPs, latency, memory, and real-time factor.
- Developing SilentEdge-LLM, a privacy-preserving sEMG-to-text pipeline with speaker verification, a lightweight EMG encoder, an EMG-to-LLM adapter, and quantized Llama 3.2 deployment on Jetson AGX Xavier.

SELECTED PUBLICATIONS

1. S. Thiyagarajan and D.-H. Kim, "HySV: A Hymba-Inspired Hybrid-Head Framework for Quality-Aware and Deployment-Aware Speaker Verification in Intelligent Embedded Systems," Electronics, vol. 15, no. 12, art. 2676, 2026. [Link](#)
2. S. Thiyagarajan and D.-H. Kim, "Symmetry and Asymmetry Principles in Deep Speaker Verification Systems: Balancing Robustness and Discrimination through Hybrid Neural Architectures," Symmetry, vol. 18, no. 1, art. 121, 2026. [Link](#)
3. S. Thiyagarajan and D.-H. Kim, "A Data-Oriented Conceptual Model and Hybrid TDNN-BiLSTM Framework for Context-Aware Speaker Verification in Smart Environments," 2026 IEEE International Conference on Big Data and Smart Computing (BigComp), 2026. [Link](#)
4. S. Thiyagarajan, H. Shin, and D.-H. Kim, "A Deep Neural Network Approach for Speaker Verification: Hybrid CNN and Multi-Head Attention with Triplet Loss Framework," Korean Institute of Information Scientists and Engineers Conference Proceedings, pp. 313-315, 2025.
5. G. Singh and S. Thiyagarajan, "The Multiple Approaches for Drug-Drug Interaction Extraction Using Machine Learning and Transformer-Based Model," bioRxiv preprint, 2025. [Link](#)

CURRENT RESEARCH PROJECT

SilentEdge-LLM Privacy-Preserving Silent Speech Recognition	In Progress
- Processes multi-channel surface electromyography signals through preprocessing, speaker verification, a lightweight EMG encoder, and an EMG-to-LLM adapter for silent-speech transcription. - Targets on-device inference using quantized Llama 3.2 on NVIDIA Jetson AGX Xavier, with evaluation across desktop and edge hardware for latency, throughput, P95 latency, memory, and real-time factor. - Prioritizes model development and edge optimization while keeping sensitive biosignals and outputs local to the device.	

TEACHING EXPERIENCE

Laboratory Assistant, Computer Programming Python <i>Inha University</i>	Mar 2025 & Sep 2025
Support a Python programming course covering Python fundamentals, control flow, functions, data structures, files, object-oriented programming, and practical exercises.	
Laboratory Assistant, Object-Oriented Programming	Mar 2026 & Sep 2026
Prepared and reviewed beginner-to-intermediate programming exercises and supported students during laboratory practice.	

EARLIER RESEARCH AND PROJECTS

Stress Detection from Physiological Signals
Compared LSTM and XGBoost approaches for multimodal physiological-signal classification.
Drug-Drug Interaction Extraction
Investigated classical machine-learning and transformer-based methods for extracting interactions from biomedical text.
Portable Banana Embryo Detection Device
Developed a handheld Arduino and computer-vision prototype under Anna University's CSRC Student Innovative Project support scheme.

INDUSTRY EXPERIENCE

Advanced AI Intern, The Korean Academy	Jun 2024 - Aug 2024
Worked on a computer-vision system for identifying flagged vehicles from license-plate imagery.	
Java Full-Stack Development Intern, Eazy Bytes	Jul 2024
Completed practical training in Java-based application development and web technologies.	

TECHNICAL SKILLS

Programming: Python, C, C++, Java, SQL, HTML, CSS
AI and ML: PyTorch, deep learning, machine learning, CNN, TDNN, BiLSTM, Transformers, multi-head attention, state-space models, multimodal fusion
Speech and Biosignals: speaker verification, audio-visual recognition, sEMG preprocessing, feature extraction, EER, minDCF
Edge Deployment: NVIDIA Jetson AGX Xavier, JetPack, CUDA, TensorRT, Ollama, model quantization, latency and memory benchmarking
Tools: Hugging Face Transformers, OpenCV, MATLAB, Arduino IDE

SCHOLARSHIPS LEADERSHIP AND CERTIFICATIONS

Jungseok International Scholarship (100%), Inha University	2025 - Present
NCC C Certificate, Ministry of Defence, Government of India	2019 - 2022
Combined Annual Training Camp, National Cadet Corps	2022
React JS Workshop, Madras Institute of Technology, Anna University	2022